

LITCHFIELD, MN, USA

WMO: 726583

Lat: 45.147N

Lon: 94.507W

Elev: 1140

StdP: 14.10

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04941

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-8.8	-20.3	1.9	-12.7	-16.6	2.3	-8.9	25.5	11.9	23.1	14.6	6.4	310	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	89.8	74.0	86.1	72.4	82.5	70.0	77.6	85.4	75.4	82.3	73.5	80.0	7.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	137.3	82.2	72.9	127.5	79.8	71.9	123.0	78.5	41.9	84.3	39.7	82.7	37.9	80.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	19.7	17.6	-19.7	94.4	5.9	2.8	-24.0	96.5	-27.5	98.1	-30.8	99.7	-35.1	101.8
			-20.3	80.9	5.7	3.1	-24.4	83.2	-27.8	85.0	-31.0	86.7	-35.1	89.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.0	13.7	17.4	30.6	44.8	57.7	67.8	72.3	69.6	62.4	48.1	33.7	19.6
	DBStd	22.82	13.28	13.02	12.48	10.28	8.68	6.68	5.57	5.27	8.42	9.06	10.80	11.97
	HDD50	4482	1125	913	610	214	27	0	0	0	7	146	498	942
	HDD65	7950	1590	1333	1067	607	260	46	8	18	146	529	939	1407
	CDD50	2642	0	0	8	59	265	534	693	607	381	86	9	0
	CDD65	632	0	0	0	2	32	129	236	160	69	4	0	0
	CDH74	5138	0	0	1	37	347	1060	1984	1118	546	44	1	0
	CDH80	1477	0	0	0	8	107	304	633	278	141	6	0	0
Wind	WSAvg	7.3	7.6	7.8	8.1	9.0	8.3	6.9	5.6	5.3	6.2	7.7	7.5	7.3
Precipitation	PrecAvg	30.1	0.8	0.8	1.7	2.8	3.3	4.8	3.9	3.7	3.2	2.3	1.5	0.7
	PrecMax	38.4	2.0	2.0	3.5	7.0	6.7	8.0	7.4	7.6	8.2	4.8	4.0	1.7
	PrecMin	23.4	0.0	0.1	0.6	0.8	0.9	2.0	0.9	0.4	0.5	0.7	0.0	0.3
	PrecStd	4.6	0.6	0.5	0.8	1.5	1.5	1.5	1.7	1.6	1.9	1.2	1.3	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.5	52.3	66.4	79.3	90.0	91.4	93.1	90.2	88.5	79.5	66.0	48.0
		MCWB	37.7	47.8	56.4	59.8	67.6	72.7	76.9	75.3	71.6	62.2	55.4	44.9
	2%	DB	37.3	43.3	59.2	72.2	82.1	87.7	89.9	86.1	83.8	72.5	59.2	42.6
		MCWB	34.5	39.1	51.6	55.2	63.7	71.7	75.9	74.0	70.9	60.1	50.1	38.0
	5%	DB	35.6	38.6	53.7	66.1	77.5	82.9	86.3	82.3	80.9	66.4	54.0	37.1
		MCWB	34.1	36.1	48.6	52.7	61.1	68.7	73.8	71.0	67.9	56.5	47.1	35.0
	10%	DB	31.9	36.0	46.2	61.5	72.6	80.8	82.5	80.8	76.4	62.9	48.0	35.6
		MCWB	30.9	34.1	42.0	49.6	59.4	67.0	71.6	69.7	65.6	54.5	43.2	34.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	47.8	59.9	62.2	72.7	78.3	82.1	78.8	77.1	67.2	58.1	45.9
		MCDB	43.0	51.2	64.4	75.1	82.2	86.5	86.9	86.5	84.4	73.2	63.9	47.2
	2%	WB	35.7	39.4	53.5	57.7	68.0	74.7	78.4	76.2	73.4	62.5	52.1	38.2
		MCDB	37.7	42.1	58.3	68.0	76.3	82.3	86.2	82.9	79.4	68.3	57.9	40.9
	5%	WB	33.2	36.6	47.7	54.4	64.3	71.9	76.2	74.2	70.7	58.7	47.4	35.9
		MCDB	35.0	38.7	52.7	63.1	72.3	80.2	83.7	80.1	77.3	65.8	52.5	37.6
	10%	WB	30.5	33.5	42.4	51.3	61.4	69.3	74.2	72.1	67.8	55.0	43.6	33.7
		MCDB	31.4	35.7	46.8	60.3	69.6	77.0	81.0	77.6	74.7	61.3	48.4	35.2
Mean Daily Temperature Range	5% DB	MDBR	15.4	16.6	16.4	18.7	19.1	18.1	18.2	17.9	19.4	17.1	15.0	13.7
		MCDBR	17.0	17.6	23.1	27.4	26.7	21.8	21.1	20.5	23.2	24.4	22.5	16.3
		MCWBR	14.3	14.3	15.5	14.6	12.3	10.4	10.9	10.6	11.8	13.4	14.5	13.4
	5% WB	MCDBR	16.1	16.1	22.1	23.5	21.6	18.9	18.6	18.1	20.0	20.1	20.7	14.8
		MCWBR	14.0	13.3	15.5	14.5	12.4	11.3	11.3	11.2	12.5	12.9	14.6	12.3
Clear-Sky Solar Irradiance	taub	0.260	0.277	0.307	0.363	0.391	0.390	0.389	0.401	0.363	0.314	0.282	0.257	
	taud	2.437	2.382	2.416	2.327	2.304	2.356	2.365	2.320	2.401	2.534	2.524	2.460	
	Ebn at Noon	277	291	295	285	279	279	277	269	272	274	266	266	
	Edh at Noon	22	28	32	38	40	38	37	38	32	24	20	19	
All-Sky Solar Radiation	RadAvg	558	897	1197	1482	1711	1903	2044	1697	1312	839	538	397	
	RadStd	69	103	113	134	133	136	73	137	97	117	52	47	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Station Only	N/A	N/A	N/A	N/A	+1.35	N/A	N/A	N/A	N/A	N/A
	Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page